SentinelOps Weekly Progress Report

Field	Value
Course	SWENG 894 - Software Engineering Capstone Experience
Project	SentinelOps
Student	Eli Vazquez
Sprint	Sprint 1
Reporting Week	Week 1
Reporting Period	2026-05-25 to 2026-05-31
Report Date	2026-05-30
Git Repository	https://github.com/swevazquez/SentinelOpsProject
Jira Board	https://psu-capstone-sentinelops.atlassian.net/jira/software/projects/SCRUM/boards/1/backlog

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Sprint Goal

The Sprint 1 goal is to establish the first working SentinelOps foundation for a predictive maintenance workflow. The sprint focuses on representative telemetry generation, raw telemetry persistence, initial feature processing, and repeatable workflow validation.

This week focused on starting the sprint correctly by connecting Jira and GitHub traceability, implementing the first telemetry generation story, and pulling automated CI validation forward so later Sprint 1 work can be integrated safely.

Sprint Planning

Groomed Product Backlog Summary

The Sprint 1 backlog was reviewed in Jira and confirmed as the starting scope for the first vertical slice.

ID	Requirement / User Story	Priority	Estimation	Sprint	Current Status
FR-01	Telemetry Generation and Ingestion	High	5 SP	Sprint 1	Done
FR-02	Raw Telemetry Storage	High	3 SP	Sprint 1	Done
FR-03	Feature Engineering Processing	High	5 SP	Sprint 1	Done
FR-04	Workflow Orchestration	High	8 SP	Sprint 1	To Do
NFR-01	Failed Workflow Detection and Reporting	High	3 SP	Sprint 1	To Do
NFR-03	Component Responsibility Separation	High	2 SP	Sprint 1	To Do
NFR-04	Repeatable Local Execution	Medium	2 SP	Sprint 1	To Do

Sprint Backlog

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NFR-04	Repeatable Local Execution	Medium	2 SP	To Do

Definition of Done

Area	Definition of Done Criteria
Requirements	Requirement is reviewed and acceptance criteria are documented.
Design	Affected architecture, workflow, or interface design is updated.
Development	Code is implemented, committed, and aligned with the existing project structure.
Testing	Unit, smoke, or manual validation is completed as appropriate.
Integration	Changes run through the local CI validation script and GitHub Actions.
Documentation	Relevant documentation and weekly reporting evidence are updated.
Validation	Implementation is validated against acceptance criteria.

Acceptance Criteria for Sprint Backlog Items

Requirement ID	Acceptance Criteria
FR-01	Given the telemetry simulator is configured, when a telemetry generation workflow is executed, then representative telemetry data shall be produced and made available for processing.

Requirement ID	Acceptance Criteria
FR-02	Given telemetry data is generated or ingested, when the ingestion workflow completes, then the telemetry data shall be persisted in the configured storage location.
FR-03	Given raw telemetry data exists, when the feature engineering workflow executes, then processed feature sets shall be generated and stored for predictive scoring.
FR-04	Given workflow definitions are configured, when a workflow is triggered, then telemetry ingestion, processing, scoring, and reporting tasks shall execute in the defined sequence.
NFR-01	Given workflow execution fails, when workflow status or logs are reviewed, then the failure shall be detectable and reportable.
NFR-03	Given Sprint 1 implementation work is complete, when the repository structure is reviewed, then API, orchestration, processing, simulator, dashboard, and agent responsibilities shall remain separated.
NFR-04	Given a clean local checkout, when setup and validation commands are run, then the Sprint 1 workflow shall execute repeatably with documented commands.

Backlog Grooming

Backlog Changes This Week

Change Type	Requirement ID	Description	Rationale	Impact
Moved earlier	NFR-04	Automated CI validation was pulled forward into Sprint 1 planning.	Early telemetry, storage, feature, and workflow work needs repeatable verification before additional services are added.	Adds pipeline work to the first sprint foundation and provides testing evidence for weekly reporting.
Estimated	NFR-01 / NFR-03 / NFR-04	Sprint 1 NFR stories were assigned story point estimates in Jira.	The NFRs represent real sprint work and should be visible in sprint capacity and burndown tracking.	Sprint 1 estimated scope increased from 21 SP to 28 SP, with 15 SP remaining after Week 1 progress.

Backlog Grooming Rationale

Automated testing and CI validation were pulled forward because the first sprint depends on several connected workflow steps. Establishing branch and pull-request validation now reduces integration risk before additional API, dashboard, prediction, and agent components are added. The Sprint 1 NFRs were also estimated so the burndown reflects the full sprint scope instead of only functional stories.

Source Code Development

Summary of Contributions

Development this week focused on project traceability, telemetry generation, and automated validation.

Key contributions include:

- Configured GitHub autolinks for Jira keys using the `SCRUM-` prefix.
- Added pull request traceability checks requiring Jira story keys.

- Added representative asset profile configuration for telemetry generation.
- Updated the simulator CLI to load and validate configured asset profiles.
- Added raw telemetry storage validation and persistence to the configured `data/raw/` location.
- Added feature engineering validation and persistence to the configured `data/processed/` location.
- Added CI validation for unit tests, Sprint 1 smoke workflow output counts, Airflow DAG syntax, and documentation readability.

Repository Information

Resource	Link
Git Repository	https://github.com/swevazquez/SentinelOpsProject
Jira Board	https://psu-capstone-sentinelops.atlassian.net/jira/software/projects/SCRUM/boards/1/backlog
Pull Requests	https://github.com/swevazquez/SentinelOpsProject/pull/1, https://github.com/swevazquez/SentinelOpsProject/pull/2, https://github.com/swevazquez/SentinelOpsProject/pull/3

Important Commits

Commit ID	Commit Summary	Related Requirement / User Story	Notes
3196025	Add Jira GitHub traceability workflow	NFR-04	Establishes traceability and reporting support.
d2a174c	Configure telemetry asset generation	FR-01	Adds configured representative telemetry input.
74579b8	Add CI validation pipeline	NFR-04	Adds local and GitHub Actions validation.

Commit ID	Commit Summary	Related Requirement / User Story	Notes
dfeb627	Run CI on feature branch pushes	NFR-04	Ensures branch work receives automated feedback before PR merge.
29b44e1	Persist raw telemetry output	FR-02	Adds validated raw telemetry persistence and updates workflow callers.
cd9085b	Validate and persist engineered features	FR-03	Adds raw input validation, processed feature persistence, and feature contract expansion.
c81a36a	Update Week 1 sprint report burndown	Sprint reporting	Adds the Week 1 burndown chart based on Jira story status.

Burndown Summary

Sprint 1 is underway. `FR-01`, `FR-02`, and `FR-03` are complete. The project foundation now includes automated validation, raw telemetry persistence, and processed feature persistence. Remaining estimated work includes `FR-04` workflow orchestration, failed workflow detection and reporting, component responsibility verification, and closure of repeatable local execution evidence.

Metric	Value
Sprint Total Estimated Effort	28 story points
Completed Effort	13 story points
Remaining Effort	15 story points
Sprint Status	On Track

Burndown Chart

Jira issue status and story point estimates were available, but a Jira-generated burndown chart was not available through the current integration tools. The chart below was generated from the Sprint 1 Jira status after `SCRUM-1`, `SCRUM-2`, and `SCRUM-3` were marked done and after Sprint 1 NFR stories were estimated.

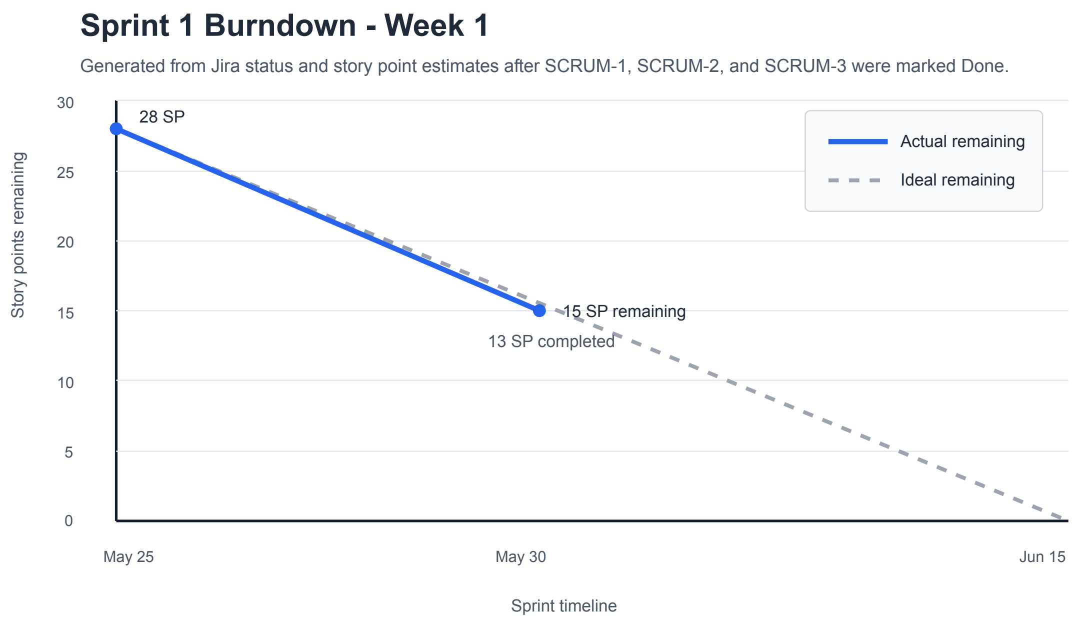

Figure 1: Sprint 1 Week 1 Burndown

Software Testing

Testing Overview

Automated testing was expanded and connected to CI. The project now has unit coverage for telemetry generation and asset profile validation, plus a smoke workflow that generates raw telemetry and processed features.

Requirement-to-Test Traceability Matrix

Requirement ID	Test Case ID	Test Type	Test Objective	Status
FR-01	TC-FR01-01	Unit	Validate telemetry generation creates hourly rows for each representative asset.	Passed
FR-01	TC-FR01-02	Unit	Validate asset profile configuration is loaded and used by telemetry generation.	Passed
FR-01	TC-FR01-03	Unit	Validate invalid asset risk configuration is rejected.	Passed
FR-02	TC-FR02-01	Unit	Validate raw telemetry is persisted to the configured storage location.	Passed
FR-02	TC-FR02-02	Unit	Validate raw telemetry storage rejects empty or inconsistent run data.	Passed
FR-03	TC-FR03-01	Unit	Validate raw telemetry is grouped into asset-level feature rows.	Passed
FR-03	TC-FR03-02	Unit	Validate feature rows are persisted to the configured processed storage location.	Passed

Requirement ID	Test Case ID	Test Type	Test Objective	Status
FR-03	TC-FR03-03	Unit	Validate missing or empty raw telemetry input is rejected.	Passed
NFR-04	TC-NFR04-01	CI Smoke	Validate the Sprint 1 local workflow generates expected raw and processed outputs.	Passed
NFR-04	TC-NFR04-02	CI	Validate GitHub Actions runs on feature branch pushes and PRs.	Passed

Test Case Specifications

TC-FR01-01 - Telemetry Generation Produces Representative Rows

Field	Description
Related Requirement	FR-01
Test Type	Unit
Preconditions	Telemetry simulator is configured with representative assets.
Test Steps	Run `python3 -m unittest discover -s tests`.
Expected Result	Telemetry rows are generated for each asset and hour with the expected fields.
Actual Result	Passed.
Status	Passed

TC-NFR04-01 - Sprint 1 CI Smoke Workflow

Field	Description
Related Requirement	NFR-04
Test Type	CI Smoke

Field	Description
Preconditions	Repository checkout and Python 3.12 are available.
Test Steps	Run `./scripts/check-ci.sh`.
Expected Result	Unit tests pass, smoke data is generated, expected output counts match, and DAG syntax is valid.
Actual Result	Passed locally and in GitHub Actions.
Status	Passed

TC-FR02-01 - Raw Telemetry Storage

Field	Description
Related Requirement	FR-02
Test Type	Unit
Preconditions	Generated telemetry rows exist for a single run.
Test Steps	Persist rows with `persist_raw_telemetry`.
Expected Result	The system creates `telemetry_<run_id>.csv` in the configured storage directory and preserves the raw telemetry schema.
Actual Result	Passed.
Status	Passed

TC-FR03-01 - Feature Engineering Processing

Field	Description
Related Requirement	FR-03
Test Type	Unit
Preconditions	Raw telemetry CSV contains the required telemetry fields.
Test Steps	Run `engineer_features` against raw telemetry input.
Expected Result	The system produces one processed feature row per `run_id` and `asset_id`, including timestamp bounds, aggregate sensor values, runtime bounds, and failure observation.

Field	Description
Actual Result	Passed.
Status	Passed

Testing Summary

Local validation passed with `./scripts/check-ci.sh`. GitHub Actions passed on PR #1, PR #2, and PR #3.

Risks, Roadblocks, and Mitigation

Risk / Roadblock	Impact	Mitigation / Next Step
Sprint 1 spans several connected workflow steps.	Integration issues may appear when telemetry, storage, features, and orchestration are combined.	Keep CI smoke validation active and extend it as each Sprint 1 story lands.

Plan for Next Week

Next week's work will focus on continuing the Sprint 1 vertical slice.

- Proceed to `SCRUM-4` workflow orchestration.
- Add workflow failure visibility as part of `NFR-01`.
- Expand validation evidence as storage and feature processing are refined.
- Keep Jira, GitHub PRs, commits, and weekly reports aligned.

Overall Sprint Assessment

Sprint progress is currently on track. The first three functional stories are complete, the codebase has a CI foundation, and the project now has stronger traceability between Jira, GitHub, tests, and weekly reporting. The main focus remains completing the Sprint 1 vertical slice and closing the supporting NFR work without expanding scope.